Liquid Methane (CH_4) & Liquid Oxygen (LOX) Densification Protocol

1. Propellant Subcooling

Liquid methane is densified to 90 K and liquid oxygen to 66 K using closed-cycle neon cryocoolers, increasing bulk propellant loading density by 8.4 percent inside the Earth Return Vehicle tanks.

2. Boil-Off Zero-Vent Storage

Cryogenic tanks incorporate 40-layer double-aluminized Mylar multi-layer insulation (MLI) and pulse-tube active re-liquefaction to eliminate venting losses over 500-sol dwell periods.

3. Transfer Line Purge

All cryogenic propellant transfer lines are vacuum-jacketed and pre-chilled with gaseous helium before liquid propellant transit to prevent cavitation shockwaves in turbopumps.